The Latency Race: Registers vs. RAM

Quiz Review & Memory Hierarchy

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The Memory Hierarchy & I/O Overhead

Storage Capacity Increases

CPU Registers (Accumulator)
$t_{reg} \approx 1$ **ns**

CPU Cache (L1/L2/L3)
$t_{cache} \approx 5$ **ns**

Main Memory (RAM)
$t_{ram} \approx 50$ **ns**

Secondary Storage (SSD/HDD)
$t_{hdd} \approx 1\mu s - 10$ **ms**

Speed & Cost Decrease

Total Execution Time via Memory Access:

$$T_{total} = (t_{reg} \times n_{reg}) + (t_{ram} \times n_{ram}) + (t_{hdd} \times n_{hdd})$$

Where t is latency and n is the number of operations at that level.

Quiz Review: Question 1

In computer architecture, which storage location is the fastest for the CPU to access?

A) The external hard drive

B) The RAM (Mailboxes)

C) A CPU Register (like the Accumulator)

D) Cloud storage

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C) A CPU Register (like the Accumulator)
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Answer: C

Registers are built directly into the CPU, meaning the data doesn't have to travel across the motherboard bus.

Quiz Review: Question 2

In the context of the Little Man Computer, what happens during a "STO" instruction?

A) Data is moved from the slow RAM into the fast Accumulator.

B) Data is moved from the fast Accumulator across the bus to the slower RAM mailboxes.

C) The program stops executing completely.

D) The computer adds a number to the Accumulator.

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Answer: B

"Store" pushes data out of the CPU and into main memory, incurring a latency penalty.

Quiz Review: Question 3

Why did Program A (The RAM Hog) take significantly more instruction cycles to complete the same task as Program B?

A) Because it had to perform complex multiplication.
B) Because it used unnecessary STO and LDA instructions to move data in and out of RAM.
C) Because the user typed the inputs in too slowly.
D) Because Program A had more INP instructions.

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Answer: B

Every trip to RAM costs time. Program A wasted cycles moving data it was about to use immediately anyway.

Quiz Review: Question 4

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A) The von Neumann Bottleneck
B) The Turing Paradox
C) The Silicon Limit
D) The Accumulator Overflow

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Answer: A

The Von Neumann Bottleneck refers to the limited throughput (data transfer rate) between the CPU and memory.

Quiz Review: Question 5

Why was Program B able to completely avoid using the "STO" (Store) command?

A) Because the program didn't actually do any math.
B) Because the data was processed and outputted immediately while it was still sitting in the Accumulator.
C) Because the "OUT" command automatically stores data in RAM.
D) Because the RAM was full.

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Answer: B

If you don't need to save a variable for later, process it in the register immediately!

Quiz Review: Question 6

If fetching data from RAM takes 50ns, and accessing the Accumulator takes 1ns, what is the best way to reduce latency?

A) Use the STO command as often as possible.

B) Minimize STO and LDA commands by doing math on data while it is still in the Accumulator.

C) Add more INP commands to slow the user down.

D) Use the HLT command earlier in the program.

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Answer: B

Minimizing n_{ram} in our equation $T_{total} = (t_{reg} \times n_{reg}) + (t_{ram} \times n_{ram})$ drastically lowers total time.

Quiz Review: Question 7

What happens to the existing data in the Accumulator the moment a new "INP" command is executed?

A) It is automatically saved to mailbox 99.
B) It is pushed down into a secondary register.
C) It is completely overwritten and erased by the new input.
D) It is added to the new input automatically.

Quiz Review: Question 7

What happens to the existing data in the Accumulator the moment a new "INP" command is executed?

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C) It is completely overwritten and erased by the new input.
D) It is added to the new input automatically.

Answer: C

The Accumulator holds exactly one value. New inputs destroy old data unless you intentionally Store it first.

Quiz Review: Question 8

How many memory access instructions (STO and LDA) did Program A execute compared to Program B?

A) Program A executed 6, while Program B executed 0.
B) Both programs executed the exact same number.
C) Program A executed 0, while Program B executed 6.
D) Program A executed 3, while Program B executed 3.

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C) Program A executed 0, while Program B executed 6.

D) Program A executed 3, while Program B executed 3.

Answer: A

Program A had 3 STO and 3 LDA commands (6 total). Program B handled everything strictly in the CPU.

Quiz Review: Question 9

Optimizing code to behave like Program B helps the CPU utilize what hardware component efficiently?

A) The cooling fan
B) The CPU Cache and internal registers
C) The solid-state drive (SSD)
D) The graphics card (GPU)

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Answer: B

Modern compilers do exactly this: they try to keep variables inside fast CPU Cache and Registers to avoid hitting the slower RAM.

Quiz Review: Question 10

If a program requires you to use a user's inputted number multiple times throughout a long program, what must you do despite the latency cost?

A) Output it immediately.

B) Use the STO command to save it to RAM so it isn't overwritten.

C) Ask the user to input the number again every time.

D) Use the HLT command.

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Answer: B

Latency is the cost of doing business when we need persistent variables. We accept the t_{ram} delay to ensure data isn't lost!

Great Job! Any questions on Memory Hierarchy?